



Solutions:

1. A. 0

Displacement is the change in position. So if you travel a certain distance, but then go back to the initial position, then there is no change in position, hence the displacement is zero.

Reference: GEAS 02-01 – Scalars and Vectors

2. A. 0

The displacement of the athlete here is 0.

$$v_{\text{average}} = \frac{\Delta x}{\Delta t} = \frac{0}{80} = 0$$

Reference: GEAS 02-01 – Scalars and Vectors

3. B. 6.3 m/s

The distance travelled of the athlete here is 500 m.

$$\text{speed}_{\text{average}} = \frac{\text{distance}}{\Delta t} = \frac{500}{80} = 6.3 \frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-01 – Scalars and Vectors

4. B. 45 m

Use the Law of Conservation of Energy:

$$PE_1 + KE_1 = PE_2 + KE_2 + \text{loss}$$

$$PE_1 + \cancel{KE_1} = \cancel{PE_2} + KE_2 + \cancel{\text{loss}}$$

$$mgh_1 = \frac{1}{2}mv_2^2$$

$$(9.81)h_1 = \frac{1}{2}(30)^2$$

$$h_1 = 46 \text{ m}$$

Reference: GEAS 02-05 – Conservation of Energy

5. A. 0

Since the object will move at “constant velocity”, meaning it’s not accelerating, there is no net force required (Law of Inertia)

Reference: GEAS 02-02 – Forces

6. B. 3 m/s²

$$F_{\text{net}} = ma$$

$$-8\text{N} + 20\text{N} = (4 \text{ kg})a$$

$$a = 3 \frac{\text{m}}{\text{s}^2}$$

Reference: GEAS 02-02 – Forces

7. C. the woman and bus experience the same force

By Law of Reaction, both of them will exert equal but opposite forces to each other.

Reference: GEAS 02-02 – Forces

8. A. $\sqrt{\frac{GM}{R}}$

$\sqrt{\frac{GM}{R}}$ is the formula of orbital speed as derived in GEAS 02-03.

Reference: GEAS 02-03 – Law of Gravitation

9. C. 4.45 N

Conversion factor : 1 lbf = 4.45 N

10. D. The period increases by a factor of square root of 8

The period of the satellite is the period it takes to make one complete revolution:

$$T = \frac{2\pi R}{v} = \frac{2\pi R}{\sqrt{\frac{GM}{R}}} = 2\pi R \sqrt{\frac{R}{GM}}$$

$$T' = 2\pi(4R) \sqrt{\frac{4R}{GM}} = 8 \left(2\pi R \sqrt{\frac{R}{GM}} \right)$$

$$T' = 8T$$

Reference: GEAS 02-03 – Law of Gravitation

11. D. none of the above

A is not necessarily true. The object could be at rest BUT it is not necessarily at rest.

B and C represent objects that are accelerating (changing velocity)

So the answer should be D.

Reference: GEAS 02-02 – Forces

12. A. decreases by a factor of 4

According to the Law of Gravitation, the gravitational force is inversely proportional to the square of distance. So if you double the distance, the magnitude of the force decreases by a factor of 2² or 4.

Reference: GEAS 02-03 – Law of Gravitation

13. B. 4.5 m/s²

$$g = \frac{GM}{R^2} = \frac{G \left(\frac{1}{500} (6 \times 10^{24} \text{ kg}) \right)}{\left(\frac{1}{15} (6400 \times 10^3 \text{ m}) \right)^2}$$

$$g = 4.5 \frac{\text{m}}{\text{s}^2}$$

Reference: GEAS 02-03 – Law of Gravitation

14. D. its mass is m and its weight is w/9

The mass m of an object will never change regardless of the location.

At an altitude h = 2R,

$$g' = \frac{GM}{(R+h)^2} = \frac{GM}{(R+2R)^2} = \frac{1}{9} \frac{GM}{R^2}$$

$$g' = \frac{1}{9} g$$

So if the acceleration due to gravity is 1/9 of the initial value, then the weight must be 1/9 of the initial value.

Reference: GEAS 02-03 – Law of Gravitation



15. D. 60 J

Using the Work-Energy Theorem:

$$W = \Delta PE$$

$$W = mgh_f - mgh_i$$

$$W = (2\text{kg})\left(9.81\frac{\text{m}}{\text{s}^2}\right)(3\text{m}) - 0 = 59\text{J}$$

Reference: GEAS 02-05 –
Conservation of Energy

16. D. 400 W

$$P = \frac{W}{t} = F_{\text{ave}} v_{\text{ave}} = (200\text{N})\left(2\frac{\text{m}}{\text{s}}\right)$$

$$P = 400\text{W}$$

Reference: GEAS 02-04 – Work,
Energy, Power

17. B. 6 m/s

Use the Law of Conservation of
Energy:

$$PE_1 + KE_1 = PE_2 + KE_2 + \text{loss}$$

$$PE_1 + \cancel{KE_1} = \cancel{PE_2} + KE_2 + \cancel{\text{loss}}$$

$$mgh_1 = \frac{1}{2}mv_2^2$$

$$(9.81)(1.8\text{m}) = \frac{1}{2}v_2^2$$

$$v_2 = 6\frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-05 –
Conservation of Energy

18. C. 30 m/s

Use the Law of Conservation of Energy:

$$PE_1 + KE_1 = PE_2 + KE_2 + \text{loss}$$

$$PE_1 + \cancel{KE_1} = \cancel{PE_2} + KE_2 + \text{loss}$$

$$mgh_1 = \frac{1}{2}mv_2^2 + fd$$

$$(40)(9.81)(50) = \frac{1}{2}(40)v_2^2 + (40\text{N})(50\text{m})$$

$$v_2 = 30\frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-05 –
Conservation of Energy

19. B. 2 cm

Using the Work-Energy Theorem:

$$W = \Delta KE$$

$$Fd = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

$$(25\text{N})d = \frac{1}{2}(0.25\text{kg})\left(2\frac{\text{m}}{\text{s}}\right)^2 - 0$$

$$d = 2\text{cm}$$

Reference: GEAS 02-05 –
Conservation of Energy

20. A. 4 m/s

Use the Law of Conservation of
Energy:

$$PE_1 + KE_1 = PE_2 + KE_2 + \text{loss}$$

$$PE_1 + \cancel{KE_1} = \cancel{PE_2} + KE_2 + \text{loss}$$

$$mgh_1 = \frac{1}{2}mv_2^2 + fd$$

$$(3)(9.81)(4\text{m}) = \frac{1}{2}(3)v_2^2 + (16\text{N})(6\text{m})$$

$$v_2 = 4\frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-05 –
Conservation of Energy

21. D. 20 m/s

Use the Law of Conservation of
Energy:

$$PE_1 + KE_1 = PE_2 + KE_2 + \text{loss}$$

$$PE_1 + \cancel{KE_1} = \cancel{PE_2} + KE_2 + \text{loss}$$

$$mgh_1 = \frac{1}{2}mv_2^2 + fd$$

$$(4)(9.81)(40) = \frac{1}{2}(4)v_2^2 + (20\text{N})(40\text{m})$$

$$v_2 = 20\frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-05 –
Conservation of Energy

22. A. 0

$$W = Fd \cos \theta = (20)(4) \cos 90^\circ = 0$$

Reference: GEAS 02-04 – Work,
Energy, Power

23. A. + 1.3 m/s

Use the Law of Conservation of
Momentum:

$$(m_1 v_1 + m_2 v_2)_{bc} = (m_1 v_1 + m_2 v_2)_{ac}$$

$$((0.5)(+4) + (0.3)(-2))_{bc} = ((0.5)(+2) + (0.3)v_2)_{ac}$$

$$v_{2ac} = 1.3\frac{\text{m}}{\text{s}}$$

Note: Remember that velocity is a
vector quantity. The velocities
before collisions should have
different signs because initially the
two objects are moving in different
directions (“the collision is head-
on”)

Reference: GEAS 02-07 – Collisions

24. A. 1.8 m/s

Use the Law of Conservation of
Momentum:

$$(m_1 v_1 + m_2 v_2)_{bc} = (m_1 v_1 + m_2 v_2)_{ac}$$

$$(m_1 v_1 + m_2 v_2)_{bc} = (m_1 + m_2) v_{ac}$$

$$((0.5)(4) + (0.3)(-2))_{bc} = (0.5 + 0.3) v_{ac}$$

$$v_{ac} = 1.8\frac{\text{m}}{\text{s}}$$

Note: Remember that velocity is a
vector quantity. The velocities
before collisions should have
different signs because initially the
two objects are moving in different
directions (“the collision is head-
on”)

Reference: GEAS 02-07 – Collisions

25. A. 2 m/s

Use the Law of Conservation of
Momentum:

$$(m_1 v_1 + m_2 v_2)_{bc} = (m_1 v_1 + m_2 v_2)_{ac}$$

$$(Mv + mv)_{bc} = (Mv_1 + mv_2)_{ac}$$

$$((4)(6) + 0)_{bc} = (4v_1 + 6(4v_1))_{ac}$$

$$v_{1ac} = 2\frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-07 – Collisions



26. C. Kinetic energy is never conserved

A and B are both incorrect since the total momentum is always conserved regardless of the type of collision.

D is wrong because kinetic energy is never conserved in inelastic collisions.

Reference: GEAS 02-07 – Collisions

27. C. the collision was not elastic and total linear momentum was conserved

Solution:

Since some kinetic energy was lost during the collision, the collision should be inelastic. That means A and B are incorrect. Also D is incorrect because in all collisions, the total momentum should be conserved.

Reference: GEAS 02-07 – Collisions

28. A. Elastic

Use the Law of Conservation of Momentum:

$$\begin{aligned}(m_1 v_1 + m_2 v_2)_{bc} &= (m_1 v_1 + m_2 v_2)_{ac} \\ (MV + mv)_{bc} &= (Mv_1 + mv_2)_{ac} \\ ((4)(6) + 0)_{bc} &= (4v_1 + 6(4v_1))_{ac} \\ v_{1ac} &= 2 \frac{m}{s}\end{aligned}$$

Solve now for the coefficient of restitution:

$$\begin{aligned}e &= - \frac{\text{relative velocity after collision}}{\text{relative velocity before collision}} \\ e &= - \frac{8 - 2}{0 - 6} = 1\end{aligned}$$

Since the coefficient of restitution is 1, then the collision must be elastic.

Reference: GEAS 02-07 – Collisions

29. B. 1.6 kN

Solve first for the velocity of the man as he reaches the ground/target.

Use conservation of energy:

$$PE_1 + KE_1 = PE_2 + KE_2 + \text{loss}$$

$$PE_1 + \cancel{KE_1} = \cancel{PE_2} + KE_2 + \cancel{\text{loss}}$$

$$mgh_1 = \frac{1}{2} m v_2^2$$

$$(9.81)(45m) = \frac{1}{2} v_2^2$$

$$v_2 = 29.7 \frac{m}{s}$$

Use impulse and momentum:

$$Ft = Dmv$$

$$F(1.5 s) = 0 - (80kg)(-29.7)$$

$$F = 1.6kN$$

Reference: GEAS 02-05 – Conservation of Energy and GEAS 02-06 – Impulse and Momentum

30. D. II and III only

In SHM, the direction of the acceleration is changing hence I is incorrect. II and III are both properties of SHM.

Reference: GEAS 02-08 – Simple Harmonic Motion

31. B. 5 cm

Using Hooke's Law where the force F = weight of the block:

$$F = kx$$

$$W = kx$$

$$(1.5)(9.81) = 300x$$

$$x = 5cm$$

Reference: GEAS 02-08 – Simple Harmonic Motion

32. D. The period will not change

If both the mass of the block (m) and the spring constant (k) will double, then there is no change in the period.

Reference: GEAS 02-08 – Simple Harmonic Motion

33. A. The frequency increases by a factor of 2

The frequency of oscillation of a mass-spring system is given by the

$$\text{formula: } f = \frac{1}{2\pi} \sqrt{\frac{k}{m}}$$

So that means if the mass is replaced with a new one that has ¼ the original value, the frequency increases by a factor of 2.

Reference: GEAS 02-08 – Simple Harmonic Motion

34. B. 2 Hz

$$T = 0.25 s \cdot 2 = 0.5 s$$

$$f = \frac{1}{T} = 2Hz$$

Reference: GEAS 02-08 – Simple Harmonic Motion

35. C. the speed is the minimum

In SHM, the acceleration is always maximum when the speed is minimum

Reference: GEAS 02-08 – Simple Harmonic Motion

36. B. – 30 J

Using the Work-Energy Theorem:

$$W = \Delta KE = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$W = 0 - 30J = -30J$$

Reference: GEAS 02-04 – Work, Energy, Power

37. D. Perfectly Inelastic

In collisions, a perfectly inelastic collision is one where the objects stick together after colliding.

Reference: GEAS 02-07 – Collisions

38. C. 68 kg

$$\text{Conversion: } 1 kg = 2.2 lb$$

39. C. Kepler's Third Law

Kepler's Three Laws of Planetary Motion:



1 – The path of planets around the sun are elliptical with the sun at one focus of the ellipse (Law of Ellipses)

2 – An imaginary line drawn from the center of the sun to the center of the planet will sweep out equal areas in equal intervals of time (Law of Equal Areas)

3 – The ratio of the square of the periods of any two planets is equal to the ratio of the cubes of their average distances from the sun (Law of Harmonies)

40. C. independent

In SHM, the frequency of oscillation has nothing to do with the amplitude of the oscillation

Reference: GEAS 02-08 – Simple Harmonic Motion

41. A. First Law of Motion

The First Law of Motion or the Law of Inertia states that objects subjected to zero net force will never accelerate.

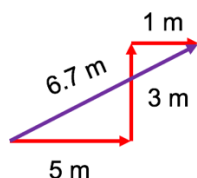
Reference: GEAS 02-02 – Forces

42. D. - 6 m/s²

$$a = \frac{\Delta v}{t} = \frac{20 \frac{\text{m}}{\text{s}} - 32 \frac{\text{m}}{\text{s}}}{2\text{s}} = -6 \frac{\text{m}}{\text{s}^2}$$

Reference: GEAS 02-01 – Scalars and Vectors

43. C. 6.7 m



Reference: GEAS 02-01 – Scalars and Vectors

44. A. 0.34 m/s

$$v_{\text{average}} = \frac{\Delta x}{t} = \frac{6.7 \text{ m}}{20 \text{ s}} = 0.34 \frac{\text{m}}{\text{s}}$$

Reference: GEAS 02-01 – Scalars and Vectors

45. B. 8t - 6

$$v_{\text{ins}} = \frac{dx}{dt} = 8t - 6$$

Reference: GEAS 02-01 – Scalars and Vectors

46. B. 8 m/s²

$$v_{\text{ins}} = \frac{dx}{dt} = 8t - 6 \quad a = \frac{dv}{dt} = 8$$

Reference: GEAS 02-01 – Scalars and Vectors

47. D. 1.3 m/s²

$$\begin{aligned} F_{\text{net}} &= ma \\ R - W &= ma \\ 1000 - (90)(9.81) &= 90a \\ a &= 1.3 \text{ m/s}^2 \end{aligned}$$

Reference: GEAS 02-02 – Forces

48. D. - 360 N

$$\begin{aligned} F &= ma = (1800 \text{ kg}) \left(-20 \frac{\text{cm}}{\text{s}^2} \right) \frac{1 \text{ m}}{100 \text{ cm}} \\ F &= -360 \text{ N} \rightarrow \text{Ans} \end{aligned}$$

Reference: GEAS 02-02 – Forces

49. A. 12 kg

$$\begin{aligned} W &= mg \\ 174 &= (6 + m)(9.67) \\ m &= 12 \text{ kg} \end{aligned}$$

Reference: GEAS 02-02 – Forces

50. B. 14 kJ

$$KE = \frac{1}{2}mv^2$$

$$KE = \frac{1}{2} \left(225 \text{ kg} \right) \left(40 \frac{\text{km}}{\text{hr}} \times \frac{1000 \text{ m}}{3600 \text{ s}} \right)^2$$

$$KE = 13.89 \text{ kJ}$$

Reference: GEAS 02-04 – Work, Energy, Power